

EAAP-LGF: A Governance Architecture for AI in Lethal Decision Support Systems

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Abstract

AI systems are increasingly used in roles adjacent to targeting and lethal decision support. Publicly described programmes have raised governance questions that cannot be resolved by model capability alone. This paper identifies five structural governance gaps: the absence of a non-overridable prohibition layer, the absence of evidence classification, the absence of calibrated confidence thresholds with hard blocks, insufficient human authorisation structure, and inadequate post-action accountability mechanisms. It presents the Enterprise AI Architecture Pattern Lethal Systems Governance Framework (EAAP-LGF), a six-layer sequential gate architecture derived from enterprise AI governance principles and adapted to the lethal systems domain. The framework defines meaningful human control as an enforceable governance architecture in which lethal force cannot be authorised without evidence classification, calibrated confidence assessment, independent legal review, multi-authoriser approval, and immutable auditability. EAAP-LGF specifies a deterministic Prohibition Layer grounded in international humanitarian law (IHL), a four-component Compound Assessment Gate, a calibrated confidence model with a categorical prohibition on algorithm-only target identification, a multi-authoriser Human Authorisation Layer, anti-automation-bias interface requirements, and mandatory post-action accountability. The paper also states the framework's limits and identifies areas requiring further legal, technical, and human-factors development.

Keywords: AI governance, lethal autonomous weapons, meaningful human control, international humanitarian law, IHL, targeting, LAWS, Project Maven, evidence classification, human-in-the-loop, prohibition layer, accountability, CCW GGE, Additional Protocol I

Authorship and intent notice. This paper is a governance constraint framework. It argues for more constraint on AI in lethal systems, not less. The author's position is that this framework is designed to constrain AI involvement, maximise human oversight, and ensure accountability, not to legitimise autonomous lethal AI systems. The paper is not a weapons development document, a military procurement specification, or an endorsement of any specific programme.

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1. Introduction

The deployment of AI systems in roles adjacent to targeting and lethal decision support has accelerated significantly across multiple national programmes. The governance architecture of these systems, to the extent that it has been publicly described, exhibits recurring structural gaps. These are not gaps that can be addressed by improving the underlying AI model, refining rules of engagement, or training human operators to be more sceptical of AI recommendations. They are architectural gaps: absences in the structure of the system that create conditions for unlawful or erroneous lethal action regardless of the intent of those deploying the system.

The most widely cited requirement for responsible AI in lethal systems is 'meaningful human control' [1, 2, 3]. The International Committee of the Red Cross (ICRC), the United Nations Group of Governmental Experts on Lethal Autonomous Weapons Systems (CCW GGE), and a growing body of academic literature have used this phrase as a normative standard without providing a concrete technical specification of what it requires in architectural terms. The result is a gap between the principle and its implementation: states can assert compliance with meaningful human control requirements without having to demonstrate that their systems are architecturally designed to enforce it.

We define meaningful human control, for the purposes of this framework, as a governance architecture in which the human decision to authorise lethal force is made with complete evidence classification, calibrated confidence assessment, independent legal review, and non-bypassable audit, and cannot be reduced to a rubber-stamp approval regardless of operational tempo or command pressure. This definition is architectural, not normative: it specifies structural preconditions rather than a value judgement.

Project Maven, the US Department of Defense's Algorithmic Warfare Cross-Functional Team, established in April 2017, is the most publicly documented programme in which AI was applied to the analysis of surveillance data for targeting support [21]. The controversy that surrounded it, culminating in approximately 3,100 Google employees signing a public letter protesting their company's involvement and Google's subsequent withdrawal from the programme [22], raised precisely the governance questions this framework attempts to answer architecturally.

This paper presents EAAP-LGF, a governance architecture derived from a working enterprise AI governance framework, the Enterprise AI Architecture Pattern (EAAP) [4], and extended into the lethal systems domain. Section 12 provides a systematic mapping of the framework's requirements to their IHL legal basis. We state explicitly in Section 11 what the framework cannot do.

1.1 Authorship and Intent

This paper is a governance constraint framework. Nothing in this paper should be read as a claim that AI can be made safe for autonomous lethal decision-making. The author's position is the opposite: this framework is designed to constrain AI involvement, maximise human oversight, and ensure accountability. It is not a weapons development document, a military procurement specification, or an endorsement of any specific programme.

1.2 Legal Review Status

This paper has not been peer reviewed and has not received formal legal validation. The IHL mappings and Article 36 discussion are offered as a governance architecture interpretation for scholarly and policy review. They must not be treated as legal advice, an operational compliance determination, or evidence that any specific system complies with IHL. Any operational, policy, procurement, or implementation use would require review by qualified IHL counsel, competent military authority, and independent technical assessors in the relevant jurisdiction and operational context.

2. Background and Related Work

2.1 International Frameworks on LAWS

The CCW Group of Governmental Experts on LAWS has been meeting since 2014 to discuss a potential international instrument governing lethal autonomous weapons systems [5]. The GGE has produced guiding principles [6] that include the requirement for human judgement over the use of force, compliance with IHL, and individual

accountability. These principles have not been translated into a legally binding instrument, and they have not been operationalised as technical requirements.

The ICRC has called for new international rules on autonomous weapons [7], specifically identifying the need for 'meaningful human control' over life-and-death decisions. The 2021 ICRC position paper identifies the requirement that humans must make context-specific judgements about the use of force, assess compliance with IHL, and bear individual legal responsibility. It does not specify the technical architecture required to ensure these requirements are met.

Scharre [8] provides a comprehensive analysis of autonomous weapons and identifies the human-machine teaming problem as central: the risk that human operators become supervisors of AI rather than decision-makers, with automation bias systematically eroding the quality of human judgement. The present paper operationalises this concern as an architectural requirement in Layer 3.

2.2 Existing AI Governance Frameworks

Enterprise AI governance frameworks have developed significant sophistication in areas of evidence classification, confidence modelling, and human oversight requirements. The EAAP framework [4], from which EAAP-LGF is derived, implements structural governance for AI recommendations in infrastructure operations. The transfer of these principles to the lethal domain requires substantially higher thresholds, additional governance layers, and the novel introduction of a non-overridable prohibition layer.

Floridi et al. [9] propose a framework of 'soft ethics' for AI governance, identifying principles of beneficence, non-maleficence, autonomy, justice, and explicability. These principles, while normatively valuable, do not specify implementable architecture. The present paper specifies governance requirements in terms of data structures, threshold values, authorisation counts, and interface constraints that can be verified by an independent auditor.

2.3 Project Maven and the Governance Question

Project Maven, formally the Algorithmic Warfare Cross-Functional Team, was established by Deputy Secretary of Defense memorandum on April 26, 2017 [21]. Its initial and most publicly documented purpose was applying computer vision AI to the analysis of full-motion video from surveillance drones, automatically flagging objects of interest to reduce the burden on human image analysts.

In April 2018, approximately 3,100 Google employees signed an open letter [22] protesting their company's involvement in the programme, citing concern that the technology would be used to improve targeting of drone strikes. Google subsequently declined to renew the Maven contract. The controversy produced public documentation of the governance questions that Maven raised but not of the governance architecture that addressed them.

EAAP-LGF Gap	Governance question raised by Maven controversy	Publicly answered?
GAP-01 - No prohibition layer	Could the Maven system generate recommendations targeting protected persons or medical facilities? Was there a non-overridable architectural mechanism preventing such recommendations from being presented to human operators?	No. The programme's internal prohibition architecture has not been publicly described.
GAP-02 - No evidence classification	When Maven flagged a person or vehicle, was the output accompanied by a structured classification of the evidence basis which video feeds, what confidence level, from how many independent sources?	No. Evidence classification of Maven outputs has not been publicly described.
GAP-03 - No calibrated confidence with hard blocks	Were Maven's object classifications accompanied by calibrated confidence scores with defined thresholds below which the output was blocked rather than presented?	No. Maven's confidence model architecture has not been publicly described.
GAP-04 - Insufficient human authorisation	What was the human authorisation structure for actions taken on Maven-supported identification? How many	No. The human authorisation architecture has not been publicly described.

EAAP-LGF Gap	Governance question raised by Maven controversy	Publicly answered?
	independent authorisers were required? Was independent legal review structurally mandated?	
GAP-05 - Inadequate post-action accountability	Was there a systematic, automated post-action review comparing Maven's identifications against confirmed post-action identities? Were discrepancies automatically flagged?	No. Post-action review mechanisms for Maven have not been publicly described.

The purpose of this table is not to assert that Project Maven lacked these governance elements. The purpose is to illustrate that the public record does not answer the governance questions the programme's own controversy raised. Meaningful human control requires that governance architecture be verifiable, not merely asserted.

2.4 Technical Work on AI Targeting

Lewis [10] and others have argued that properly designed AI targeting support systems could reduce civilian casualties by improving target discrimination. This argument is not incompatible with EAAP-LGF, but it presupposes the existence of governance architecture that current programmes do not demonstrably have. The framework presented here is a precondition for the argument that AI can improve discrimination, not a rebuttal of it.

Human factors research on automation bias [11, 12] demonstrates that operators of semi-autonomous systems systematically over-rely on system recommendations, particularly under time pressure and cognitive load. The cognitive confirmation requirements in Layer 3 are grounded in this literature. The DoD AI Ethics Principles (2020) [23] represent the normative commitment at institutional level; EAAP-LGF represents the architectural specification required to implement those principles verifiably.

3. The Five Structural Gaps

Before presenting the governance architecture, we identify the specific structural gaps that motivate it. These gaps are derived from the publicly available description of AI targeting support programmes. We make no claims about the legality of any specific programme.

Gap	Description	Risk Consequence
GAP-01 - No prohibition layer	Current AI targeting support systems have no structurally non-overridable layer that intercepts recommendations touching absolutely prohibited categories.	A recommendation touching a protected person, protected object, or prohibited weapon may be presented to a human authoriser, creating a decision point that should not exist under IHL.
GAP-02 - No evidence classification	AI recommendations are not required to carry a structured classification of the evidence basis, which intelligence types, how reliable, how current, from how many independent sources.	The human authoriser cannot assess the reliability of what they are being asked to approve. Misidentification risk is invisible.
GAP-03 - No calibrated confidence with hard blocks	AI targeting recommendations are not accompanied by a calibrated, transparent confidence score with defined thresholds below which the recommendation is blocked, not just labelled.	Low-confidence recommendations reach human authorisers who may approve them based on apparent certainty rather than underlying evidence quality.
GAP-04 - Insufficient human authorisation	'Human in the loop' is often implemented as a single authoriser under time pressure. Legal advisor participation is not structurally required.	Human authorisation becomes a formality rather than a genuine check. Legal compliance is not independently verified on each targeting decision.
GAP-05 - Inadequate post-action accountability	Audit trails do not systematically compare pre-action assessments against post-action reality. No feedback loop identifies systematic AI error.	Accountability after an unlawful strike depends on assertion, not documentation.

4. EAAP-LGF Architecture Overview

4.1 Design Principles

EAAP-LGF is organised as a six-layer sequential gate model. Each layer must be passed before the recommendation proceeds to the next. Failure at any layer produces a hard block that cannot be bypassed by authoriser rank, operational urgency, or command authority.

The asymmetric threshold principle governs all confidence requirements. In lethal decision support, the cost of a false negative blocking a lawful recommendation is always less than the cost of a false positive permitting an unlawful or erroneous recommendation. Thresholds are therefore set conservatively.

4.2 The Six-Layer Model

Layer	Name	Function	Overridable?
-1	Prohibition Layer	Intercepts recommendations touching IHL-prohibited categories. Deterministic, tamper-resistant, cryptographically controlled, and fail-safe. Addresses GAP-01.	NO Absolute
0	Intelligence Assessment	Four-component compound gate: TIC (GAP-02), Legal Compliance, CDE, Situational Currency.	NO Hard blocks
1	Confidence & Uncertainty	Calibrated confidence scoring with thresholds. Hard blocks below minimum. Mandatory Known Unknowns disclosure. Addresses GAP-03.	NO below threshold
2	Lethality Tier Model	Risk classification: L0 (non-kinetic reversible) through L4 (mass casualty potential).	NO Effect-determined
3	Human Authorisation	Multi-authoriser with mandatory legal advisor, cognitive confirmation, approval tokens. Addresses GAP-04.	NO Minimum count
4	Audit & Accountability	Immutable pre/post-action record. Mandatory post-action review. Discrepancy detection. Addresses GAP-05.	NO Cannot be disabled

5. Layer –1: The Prohibition Layer

Layer –1 is the most important element of the framework. It intercepts any recommendation touching an IHL-prohibited category before that recommendation reaches a human authoriser. The design intent is not to make the human decision easier. It is to ensure that certain decision points do not arise at all, because IHL recognises that human judgement fails under pressure and the prohibitions exist precisely to prevent atrocities even when participants believe they have good reasons to cross the line.

The full IHL basis for each prohibition category is set out in Section 12. The prohibition layer directly addresses GAP-01: the governance question of whether a system could generate recommendations targeting protected persons would be structurally answered no such recommendation can be generated, because the deterministic prohibition enforcement mechanism intercepts and blocks it before it is presented to any human operator.

Prohibition Category	IHL Basis
Attacks on civilians	GCIV, AP I Art. 51 - presumption of civilian status
Attacks on hors de combat	GCIII, AP I Art. 41 - wounded, surrendering, captured
Attacks on medical personnel and facilities	GCI Art. 19, AP I Art. 12
Attacks on cultural property	Hague Convention 1954, AP I Art. 53
Attacks on objects indispensable to civilian survival	AP I Art. 54

Prohibition Category	IHL Basis
Attacks on works containing dangerous forces	AP I Art. 56 - dams, nuclear installations
Indiscriminate attacks	AP I Art. 51(4)
Disproportionate attacks	AP I Art. 51(5)(b) - proportionality; detailed assessment at Layer 0 LCA
Attacks using prohibited weapons	CCW, Ottawa, Oslo, Hague 1899
Perfidy	AP I Art. 37

THE PROHIBITION LAYER CANNOT BE OVERRIDDEN BY ANY PERSON OF ANY RANK OR AUTHORITY, UNDER ANY OPERATIONAL CIRCUMSTANCES. IT IS NOT CONFIGURABLE. IT CANNOT BE SUSPENDED, BYPASSED, OR DEACTIVATED. IT MUST BE DETERMINISTIC, TAMPER-RESISTANT, AND CRYPTOGRAPHICALLY CONTROLLED.

Implementation requirement: the prohibition logic must be deterministic, independently verifiable, tamper-resistant, cryptographically signed, and fail-safe. It should be implemented using appropriate technical control mechanisms, such as hardware-backed policy enforcement, cryptographic signing, separation of duties, one-way control-path constraints, and independent verification. The ruleset should require joint control by the operating authority and an independent IHL verification function so that neither party alone can weaken the prohibition rules. The failure mode must be safe: enforcement malfunction produces a hard block on all recommendations, not a bypass. The prohibition layer must not itself be an AI system. It must contain no probabilistic or learning components.

6. Layer 0: Intelligence Assessment Gate

Layer 0 implements a four-component compound assessment gate. All four components must independently pass their defined thresholds. Passing three of four is insufficient. A consolidated reference of all hard block conditions and their required resolutions is provided in Appendix A.

6.1 Assessment 1 - Target Identification Confidence (TIC)

Assessment 1 directly addresses GAP-02. Every recommendation must carry a TIC assessment with explicit evidence source classification, independence evaluation, and reliability scoring.

TIC Component	Hard Block Condition
Primary source confirmation required	No primary source → HARD BLOCK
Cross-cued confirmation from independent source of different intelligence type	No independent confirmation → HARD BLOCK for Tier L3
Identity confidence score (0-100)	Score < 80 → HARD BLOCK for Tier L3
Active spoofing indicator assessment	Active spoofing indicator detected → HARD BLOCK pending human assessment
Protected person proximity check	Protected person within exclusion zone at confidence > 60% → HARD BLOCK

6.2 Assessment 2 - Legal Compliance Assessment (LCA)

Assessment 2 is a legal assessment, not a confidence assessment. It evaluates military objective qualification (AP I Art. 52(2)), proportionality (AP I Art. 51(5)(b)), precautions in attack (AP I Art. 57), and rules of engagement compliance. For cases where proportionality is within a defined margin, or military objective qualification is based on use/purpose rather than nature, the assessment routes to mandatory human legal advisor review. For Tier L3, legal advisor

confirmation is a hard block condition. The full illustrative LCA criteria with IHL article mapping are provided in Appendix B.

6.3 Assessments 3 and 4 - CDE and Situational Currency

Assessment 3 (Collateral Damage Estimate) requires a probability distribution over casualty counts, not a point estimate. The 90th percentile of that distribution is compared to the proportionality threshold, not the median. A CDE confidence interval that extends to include protected persons at defined probability is a hard block for Tier L3.

Assessment 4 (Situational Currency Check) evaluates intelligence staleness against type-specific thresholds. Surrender indicators, including ambiguous signals, are treated as hard blocks pending human assessment of surrender status. The framework requires that any ambiguity defaults to treating the signal as a surrender indicator, not treating it as noise.

7. Layer 1: Confidence and Uncertainty

Layer 1 addresses GAP-03 by specifying a numerical confidence model with defined thresholds and non-bypassable hard blocks. The values in this paper are proposed governance thresholds for review and validation, not empirically final calibration values. Any operational implementation would require independent calibration, adversarial testing, and approval by qualified legal and technical authorities before use.

Evidence Class	Score	Notes
Confirmed - Multi-source (3+ independent types)	100	Rarely achieved operationally
Confirmed - Dual-source (2 independent, different types)	85	Standard minimum for Tier L3
Single source - High reliability, documented	65	Insufficient alone for Tier L3
Single source - Unknown or concerning reliability	40	HARD BLOCK for Tier L3
Pattern-of-life inference (AI behavioural analysis)	25	Cannot be primary identification basis
Algorithm-only identification (no human intelligence source)	10	CATEGORICALLY PROHIBITED, HARD BLOCK all tiers

ALGORITHM-ONLY IDENTIFICATION WHERE AN AI SYSTEM IDENTIFIES A TARGET WITHOUT ANY HUMAN INTELLIGENCE SOURCE INVOLVEMENT IS CATEGORICALLY INSUFFICIENT FOR LETHAL ACTION UNDER THIS FRAMEWORK. NO CONFIDENCE SCORE, AUTHORISATION LEVEL, OR OPERATIONAL CIRCUMSTANCE CHANGES THIS REQUIREMENT.

Level	Score	Actions Permitted
Confirmed	90-100	Tier L3 and below, subject to all governance gates
High	75-89	Tier L2 without restriction. Tier L3 with additional verification.
Moderate	55-74	Tier L1 and below. Tier L3: HARD BLOCK.
Low	30-54	Tier L0 only. Conditional Override triggered.
Unknown	0-29	No action. Hard block. Intelligence collection required.

Known Unknowns in a specific set of categories are categorical hard blocks at Layer 1, not confidence degraders. The full reference table is provided in Appendix A.

8. Layer 2: Lethality Tier Model

Tier	Name	Authorisers	Legal Advisor	Approval Window
L0	Non-kinetic reversible	Authentication only	No	N/A
L1	Non-kinetic potentially irreversible	1	No	60 minutes
L2	Kinetic non-lethal	2 independent	Recommended	30 minutes
L3	Kinetic potentially lethal	3 independent	MANDATORY	15 minutes
L4	Kinetic mass casualty potential	5 including political authority	MANDATORY + independent review	30 minutes

Tier classification is determined by anticipated effect type, not by the authorising commander's assessment of risk. A Tier L3 recommendation cannot be downgraded to Tier L2 by command instruction. The tier classification is determined by the system and embedded in the approval token.

9. Layer 3: Human Authorisation

9.1 Multi-Authoriser Requirement

Layer 3 directly addresses GAP-04. Multiple independent authorisers with distinct roles are required because: single-point cognitive failure cannot be excluded; automation bias is amplified in solo review; and role conflation prevents rigorous independent assessment of tactical necessity, legal compliance, and proportionality separately.

Authoriser Role	Function	Independent From	Min. for L3
Tactical Commander	Confirms military necessity and operational feasibility	All other roles	1 required
Legal Advisor	Independently confirms IHL compliance, proportionality, military objective qualification. Block is absolute.	Tactical authority system-enforced	1 required
Commanding Officer (use-of-force authority)	Confirms authorisation under applicable mandate and RoE	Verified by system	1 required
Political Authority (Tier L4 only)	Confirms political authorisation for mass casualty potential actions	Constitutional level	Required for L4

9.2 Anti-Automation-Bias Interface Requirements

Automation bias, the systematic tendency to over-rely on AI recommendations, is the primary governance failure mode in the human authorisation layer [11, 12]. The interface must actively resist it, not merely permit human review. These interface controls are intended to reduce automation bias and create auditable evidence of cognitive engagement. They do not prove that meaningful human judgement occurred in every case. Empirical human-factors validation would be required before any implementation could be treated as operationally reliable.

Requirement	Implementation
Active confirmation not passive approval	Authoriser actively states the target identification by text or voice entry. Not a confirmation click.

Requirement	Implementation
Sequential compound assessment with mandatory dwell time	'Next' disabled for minimum period based on comprehension time. Dwell time logged below-minimum times flagged in audit.
Explicit Known Unknown acknowledgement	Each Known Unknown responded to individually: acknowledged, changes assessment, or unable to assess.
Minimum deliberation time monitoring	Below-minimum time between presentation and authorisation triggers a mandatory review flag.
Dissent recording	Dedicated dissent button. Record immutable. Cannot be removed by any command authority.
Legal advisor independence	Legal advisor screen shows no tactical commander assessment. Legal block is final, with no escalation pathway.

9.3 Approval Token Architecture

The three-authoriser requirement is enforced by threshold cryptography (Shamir's Secret Sharing applied to the signing key). Three partial key shares from three distinct authoriser roles are mathematically required to produce a valid signature. Two authorisers cannot produce a complete signature regardless of their rank or authority. Approval tokens are time-limited (15 minutes for Tier L3) and cryptographically bound to a specific recommendation instance. An expired token requires a complete new authorisation cycle including a full Layer 0 reassessment.

10. Layer 4: Immutable Audit and Accountability

Layer 4 directly addresses GAP-05. Every recommendation that reaches Layer 3 regardless of whether it is ultimately authorised and executed produces an immutable pre-action audit record. The record is cryptographically chained: modification of any historical record changes its hash, which invalidates every subsequent record's chain reference. Tampering is mathematically detectable.

Review Component	Discrepancy Trigger
Target identity verification against pre-action TIC assessment	Any discrepancy → mandatory investigation AND suspension of AI recommendation capability pending review
TIC score accuracy predictive validity assessment	Pattern of TIC >80 where post-action identity incorrect → systematic model review
CDE accuracy anticipated vs. actual collateral damage	Any unanticipated civilian casualty → mandatory investigation. Systematic CDE underestimation → suspension.
Proportionality reassessment with actual effects knowledge	Actual effects disproportionate → mandatory legal review

The audit trail is maintained for a minimum of twenty years on write-once or equivalently tamper-evident storage. Destruction or modification must be technically prevented where feasible and treated as a serious legal, disciplinary, and accountability breach under applicable law. The record should be maintained in a form suitable for military justice proceedings, ICC investigations, UN fact-finding missions, parliamentary oversight processes, or other competent accountability mechanisms, subject to the applicable legal mandate.

11. What This Framework Cannot Do

Intellectual honesty about the limits of governance architecture is a precondition for the framework being taken seriously.

Limitation	Explanation
Cannot make war ethical	This framework constrains AI involvement. It cannot make war itself ethical. Atrocity can occur with perfect governance architecture if the human decisions at its centre are wrong.
Cannot prevent automation bias completely	The cognitive confirmation requirements reduce automation bias. Under sustained stress, information overload, or institutional pressure, humans will still tend to follow AI recommendations.
Cannot resolve contested IHL questions	Where IHL itself is contested novel operational contexts, new weapon types, hybrid warfare, the LCA cannot provide definitive answers. Legal uncertainty remains.
Cannot guarantee correct calibration	The confidence thresholds assume correctly calibrated models. Achieving correct calibration in adversarial, data-sparse environments is a research problem this framework presupposes but does not solve.
Cannot enforce political accountability	The framework creates an audit trail. It cannot compel states to use accountability mechanisms or substitute for political will.
Cannot address fully autonomous systems	This framework governs AI-assisted lethal decision support. It does not address fully autonomous lethal systems. A separate legally binding prohibition is required.
Cannot answer whether AI should be in targeting chains	This is a normative question that governance architecture cannot resolve.

12. Relationship to International Humanitarian Law

EAAP-LGF is designed to be consistent with established sources of international humanitarian law. This section provides a systematic mapping of the framework's requirements to their IHL legal basis. It does not constitute legal advice, a legal opinion, or an assertion that any specific system complies with IHL. All deployment, testing, and policy use must be validated by qualified IHL counsel in the relevant jurisdiction and operational context.

IHL Source	Relevance to EAAP-LGF
Geneva Conventions (1949) and Additional Protocols	Foundation of protected persons and objects in Layer –1. Principle of distinction. Precautions in attack encoded in Layer 0 Assessment 2.
AP I Articles 35-42 Basic rules	Prohibition of weapons causing superfluous injury (Layer –1). Prohibition of indiscriminate attacks (Layer –1 and Layer 0 LCA). Definition of military objectives (Layer 0 Assessment 2).
AP I Article 51 Protection of civilian population	Prohibition of indiscriminate attack (Layer –1). Prohibition of disproportionate attack (Layer –1 absolute prohibition and Layer 0 proportionality). Basis for asymmetric threshold principle in Layer 1.
AP I Article 52 General protection of civilian objects	Definition of military objectives by nature, location, purpose, or use primary encoding basis for Layer 0 Assessment 2 LCA. See Appendix B for illustrative LCA criteria.
AP I Article 57 Precautions in attack	Active duty to take all feasible precautions (Layer 0 Assessment 2). Duty to verify targets (Layer 0 Assessment 1 TIC). Duty to choose means causing least civilian harm. See Appendix B.
AP I Article 36 New means of warfare	Obligation, for states party to Additional Protocol I, to determine whether the employment of a new weapon, means, or method of warfare would be prohibited by international law applicable to that state. AI systems

IHL Source	Relevance to EAAP-LGF
	affecting targeting chains may create Article 36 review issues depending on their function, jurisdiction, operational context, and treaty obligations.
AP I Article 41 Safeguarding hors de combat	Persons who are hors de combat must not be attacked. Implemented in Layer –1 (absolute prohibition) and Layer 0 Assessment 4 (surrender indicator hard block).
Martens Clause (AP I Art. 1(2))	In cases not covered by specific treaty provisions, civilians and combatants remain under the protection of customary international law. Relevant to novel AI-generated targeting scenarios not addressed by specific IHL provisions.
CCW Group of Governmental Experts on LAWS	EAAP-LGF's meaningful human control definition is designed to be consistent with the majority CCW GGE position. The five-gap analysis is offered as a technical contribution to the GGE process.
Ottawa Treaty (1997) Anti-Personnel Mines	Categorical prohibition. Implemented in Layer –1 prohibited weapons category.
Oslo Treaty (2008) Cluster Munitions	Categorical prohibition. Implemented in Layer –1 prohibited weapons category.
UN Secretary-General's Call for LAWS Prohibition	EAAP-LGF's categorical prohibition on algorithm-only target identification is consistent with the Secretary-General's position on autonomous lethal systems.

AP I Article 36 deserves specific emphasis. It requires states party to Additional Protocol I to determine whether the employment of a new weapon, means, or method of warfare would be prohibited by international law applicable to that state. The deployment of AI systems in targeting chains may create material Article 36 review issues depending on the system's function, jurisdiction, operational context, and the state's treaty obligations. States deploying such systems without a systematic governance architecture review may therefore create a material Article 36 compliance risk. EAAP-LGF is designed to provide a governance reference architecture against which such a review could be structured and documented.

13. Governing Rules

The following governing rules apply in addition to all ten EAAP governing rules inherited from the parent framework (GR-001 through GR-010):

- LGF-GR-001 The Prohibition Layer (Layer –1) is absolutely non-overrideable. No person of any rank and no operational circumstance justifies bypassing Layer –1.
- LGF-GR-002 Algorithm-only target identification is categorically insufficient for lethal action. Every lethal recommendation must be grounded in human intelligence source confirmation.
- LGF-GR-003 A qualified legal advisor must independently confirm legal compliance for every Tier L3 and L4 action. This cannot be delegated or waived.
- LGF-GR-004 Hard block conditions are absolute. They cannot be overridden by authoriser rank, operational urgency, or command authority.
- LGF-GR-005 Every lethal action recommendation must disclose all Known Unknowns. A recommendation concealing a Known Unknown is invalid regardless of other scores.
- LGF-GR-006 Approval tokens expire and cannot be extended. An expired token requires a complete new authorisation cycle.
- LGF-GR-007 The post-action review is mandatory for all Tier L3 and L4 actions. It cannot be waived, abbreviated, or delayed.
- LGF-GR-008 Any civilian casualty not anticipated in the pre-action CDE triggers a mandatory investigation. Systematic CDE underestimation triggers suspension of AI recommendation capability.
- LGF-GR-009 The audit trail is immutable and cannot be destroyed through ordinary operational processes. Records are maintained for a minimum of 20 years. Modification or destruction must be treated as a serious legal, disciplinary, and accountability breach under applicable law.

LGF-GR-010 The AI system recommends. It does not decide. It does not execute. Decision and execution are human responsibilities.

LGF-GR-011 This framework defines minimum requirements. More constraint is always permissible. Less is not.

LGF-GR-012 The framework must be reviewed by qualified IHL counsel before any operational deployment, formal policy adoption, or proposed modification. No modification may reduce the governance requirements specified here without a documented legal and governance review.

14. Gaps Requiring Further Development

EAAP-LGF is a first-generation conceptual governance specification. The following areas require substantial further development, legal review, empirical validation, and independent technical assessment before any implementation could be considered.

Gap Area	Development Required
TIC calibration methodology	Correct calibration in adversarial, data-sparse operational environments. Research programme on model calibration with independent verification methodology.
LCA encoding	Encoding IHL proportionality and military objective criteria in computable form for contested cases. See Appendix B for illustrative criteria requiring development.
CDE modelling	Standardised, auditable computational CDE methodology. No internationally agreed standard currently exists.
Anti-automation-bias interface design	Empirical human factors research on authoriser interface design that achieves genuine cognitive engagement rather than formalised approval.
Prohibition layer independent verification	Technical methodology and organisational protocol for independent verification of prohibition layer implementations.
Cross-jurisdictional applicability	Jurisdiction-specific implementation guidance for different RoE, treaty obligations, and command structures.
Hybrid warfare contexts	Specific guidance where combatant/civilian distinction is contested. Cannot rely on clear status categorisation assumed by the current framework.

15. Comparison: Enterprise EAAP vs. EAAP-LGF

Feature	Enterprise EAAP	EAAP-LGF
Prohibition layer	None	Layer –1: deterministic, tamper-resistant, cryptographically controlled, IHL-based, non-overridable
Algorithm-only identification	Not applicable	Categorically prohibited hard block all tiers
Minimum confidence for primary action	High (≥ 80)	Confirmed (≥ 90) for Tier L3 lethal action
Human authorisers required	1 (with approval token)	3 independent (L3); 5 (L4)
Legal advisor	None	Mandatory L3/L4 block is absolute
Approval window	Up to 72 hours	15 min (L3); 30 min (L4)
Evidence classes	4	6 including algorithm-only hard block
Known Unknowns	Confidence degraders	Selected categories are hard blocks (Appendix A)
Post-action review	Not required	Mandatory L3+, automatic trigger
Audit retention	Organisational policy	20 years minimum, non-destructible
Cognitive confirmation	Not required	Mandatory anti-automation-bias interface

Feature	Enterprise EAAP	EAAP-LGF
IHL mapping	Not applicable	Full mapping Section 12 and Appendix B
Governing rules	GR-001 to GR-010	GR-001 to GR-010 + LGF-GR-001 to LGF-GR-012

16. Recommendations to Policymakers

Mandate governance architecture standards. The CCW GGE process should incorporate minimum governance architecture requirements, not just principles for AI systems involved in targeting chains. The Maven controversy demonstrated that principles without architecture are insufficient.

Require independent audit. States deploying AI in targeting chains should subject their governance architecture to independent technical and legal audit. The absence of a public governance architecture description should be treated as a public-verifiability gap, not as proof that controls do not exist.

Treat algorithm-only identification as prohibited. No lethal action should be permissible based solely on AI identification without human intelligence source confirmation. This bright line should be treaty-codified.

Invoke AP I Article 36 reviews. States party to Additional Protocol I should assess whether AI targeting-support systems affect weapons, means, or methods of warfare in a manner requiring Article 36 review, and should use EAAP-LGF or an equivalent framework as a governance reference architecture where applicable.

Mandate post-action accountability mechanisms. Every lethal action involving AI recommendation support should produce an immutable audit trail suitable for review by competent accountability mechanisms under applicable legal authority.

Invest in IHL-AI research. The calibration, LCA encoding, and CDE modelling gaps in Section 14 require sustained research investment that no single state or organisation can provide alone.

Move toward a legally binding instrument on LAWS. EAAP-LGF describes minimum requirements for AI-assisted lethal decision support. A legally binding prohibition on systems that select and engage targets without meaningful human control is required as a complement.

17. Conclusion

Project Maven, the most publicly documented AI targeting support programme, generated significant public controversy without generating a public architectural account sufficient to verify how the governance questions raised by that controversy were resolved. This paper has presented EAAP-LGF as a concrete technical specification of the governance architecture that would need to exist for those questions to be affirmatively and verifiably answered.

The framework addresses five structural gaps in currently described programmes, each mapping directly to a governance question raised by the Maven controversy that remains publicly unanswered in architectural terms. Its central contributions are: a cryptographically enforced Prohibition Layer grounded in IHL that is structurally non-overridable; the categorical prohibition on algorithm-only target identification; the multi-authoriser requirement with mandatory independent legal advisor participation; the anti-automation-bias interface requirements; and mandatory post-action review with automatic triggering and systematic discrepancy detection.

This version includes a systematic mapping of the framework's requirements to their IHL legal basis (Section 12), including the argument that AI systems affecting targeting chains may create material Article 36 review issues depending on their function, jurisdiction, operational context, and applicable treaty obligations. Known Unknowns

that function as categorical hard blocks are consolidated in Appendix A, and the illustrative LCA criteria with IHL article citations are provided in Appendix B.

The most important single requirement is stated in LGF-GR-010: the AI system recommends. It does not decide. It does not execute. Decision and execution are human responsibilities that this architecture supports, structures, constrains, and audits but does not replace.

References

- [1] International Committee of the Red Cross (ICRC). (2021). ICRC Position on Autonomous Weapons Systems. Geneva: ICRC. <https://www.icrc.org/en/document/icrc-position-autonomous-weapon-systems>
- [2] Heyns, C. (2013). Report of the Special Rapporteur on extrajudicial, summary or arbitrary executions. UN Doc A/HRC/23/47. United Nations Human Rights Council. https://www.ohchr.org/Documents/HRBodies/HRCouncil/RegularSession/Session23/A-HRC-23-47_en.pdf
- [3] Human Rights Watch and International Human Rights Clinic. (2012). Losing Humanity: The Case Against Killer Robots. New York: Human Rights Watch. <https://www.hrw.org/report/2012/11/19/losing-humanity/case-against-killer-robots>
- [4] Goulden, J. (2026). EAAP - Enterprise AI Architecture Pattern: A Governance Architecture for Deterministic Enterprise AI Systems. Zenodo. <https://doi.org/10.5281/zenodo.20085547>
- [5] United Nations Office for Disarmament Affairs. (2014-2026). Convention on Certain Conventional Weapons, Group of Governmental Experts on Lethal Autonomous Weapons Systems. Geneva. <https://disarmament.unoda.org/en/our-work/emerging-challenges/lethal-autonomous-weapon-systems>
- [6] Convention on Certain Conventional Weapons. (2019). Report of the 2019 Session of the Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons Systems. CCW/GGE.1/2019/3. Geneva. https://documents.unoda.org/wp-content/uploads/2020/09/CCW_GGE.1_2019_3_E.pdf
- [7] International Committee of the Red Cross. (2021). ICRC Calls for New Rules on Autonomous Weapons. Geneva: ICRC. <https://www.icrc.org/en/document/icrc-position-autonomous-weapon-systems>
- [8] Scharre, P. (2018). Army of None: Autonomous Weapons and the Future of War. New York: W.W. Norton & Company.
- [9] Floridi, L. et al. (2018). AI4People: An Ethical Framework for a Good AI Society. Minds and Machines, 28(4), 689-707. <https://doi.org/10.1007/s11023-018-9482-5>
- [10] Lewis, M. W. (2012). Drones and the Boundaries of the Battlefield. Texas International Law Journal, 47(2), 293-314.
- [11] Parasuraman, R., & Manzey, D. H. (2010). Complacency and Bias in Human Use of Automation. Human Factors, 52(3), 381-410. <https://doi.org/10.1177/0018720810376055>
- [12] Cummings, M. L. (2004). Automation Bias in Intelligent Time Critical Decision Support Systems. AIAA 1st Intelligent Systems Technical Conference. Chicago. <https://doi.org/10.2514/6.2004-6313>
- [13] International Committee of the Red Cross. (2019). Ethics and Autonomous Weapon Systems: An Ethical Basis for Human Control? Geneva: ICRC. <https://www.icrc.org/en/document/ethics-and-autonomous-weapon-systems-ethical-basis-human-control>
- [14] Schmitt, M. N., & Thurnher, J. S. (2013). 'Out of the Loop': Autonomous Weapon Systems and the Law of Armed Conflict. Harvard National Security Journal, 4(2), 231-281. <https://harvardnsj.org/2013/05/22/out-of-the-loop-autonomous-weapon-systems-and-the-law-of-armed-conflict/>
- [15] Roff, H. M., & Moyes, R. (2016). Technical Complexity, Artificial Intelligence and the Meaningful Human Control of Autonomous Weapons Systems. Article 36 / PAX. <https://www.article36.org/wp-content/uploads/2016/04/MHC-AI-and-AWS-FINAL.pdf>
- [16] United Nations Secretary-General. (2023). Our Common Agenda, Policy Brief 9: A New Agenda for Peace. New York: United Nations. <https://digitallibrary.un.org/record/4015374>

- [17] Additional Protocol I to the Geneva Conventions (1977). Protocol Additional to the Geneva Conventions of 12 August 1949, and Relating to the Protection of Victims of International Armed Conflicts. <https://ihl-databases.icrc.org/en/ihl-treaties/api-1977>
- [18] Convention on Certain Conventional Weapons (CCW). Protocol IV on Blinding Laser Weapons (1995). Protocol V on Explosive Remnants of War (2003). <https://digitallibrary.un.org/record/808454>
- [19] Ottawa Treaty (1997). Convention on the Prohibition of the Use, Stockpiling, Production and Transfer of Anti-Personnel Mines and on Their Destruction. <https://ihl-databases.icrc.org/en/ihl-treaties/apmbc>
- [20] Oslo Treaty (2008). Convention on Cluster Munitions. <https://ihl-databases.icrc.org/en/ihl-treaties/ccm-2008>
- [21] Office of the Deputy Secretary of Defense. (2017). Memorandum: Establishment of an Algorithmic Warfare Cross-Functional Team (Project Maven). Washington, D.C.: Department of Defense. April 26, 2017. <https://nsarchive.gwu.edu/document/18583-national-security-archive-department-defense>
- [22] Shane, S., & Wakabayashi, D. (2018). 'The Business of War': Google Employees Protest Work for the Pentagon. The New York Times. April 4, 2018. <https://www.nytimes.com/2018/04/04/technology/google-letter-ceo-pentagon-project.html>
- [23] United States Department of Defense. (2020). DoD Adopts Ethical Principles for Artificial Intelligence. DoD News. February 24, 2020. <https://www.defense.gov/News/Releases/Release/Article/2091996/dod-adopts-ethical-principles-for-artificial-intelligence/>

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This paper is a preprint. It has not been peer reviewed or legally validated. The author welcomes substantive engagement from IHL scholars, AI governance researchers, and human factors specialists. Correspondence: research@johngoulden.de

Appendix A: Known Unknowns as Hard Blocks Reference Table

The following Known Unknowns produce categorical hard blocks at Layer 1. Unlike standard Known Unknowns, which degrade the confidence score, these categories prevent the recommendation from proceeding regardless of other scores. Each requires a specific human-led resolution process before the recommendation may be resubmitted.

Known Unknown	Hard Block Tier	Resolution Required Before Resubmission
Civilian presence in target area: unconfirmed	All lethal tiers	Active assessment of civilian presence by human intelligence source with documented reliability profile.
Target identity: single source only	Tier L3 and above	Cross-cued confirmation from an independent intelligence source of a different type (e.g., HUMINT if currently SIGINT-only).
Target identity: algorithm-only (no human intelligence source)	All tiers CATEGORICAL PROHIBITION	Human intelligence source identification required. A human intelligence source must be involved. Adding more algorithmic sources does not resolve this hard block.
Protected object within effects radius: presence unassessed	All lethal tiers	Assessment of protected object type and applicability of protection. Legal advisor review required.
Rules of engagement applicability: contested	All lethal tiers	Legal advisor determination of applicable RoE in the specific operational context.
Surrender indicator: any ambiguity	All tiers	Human assessment of surrender status. If any possibility of surrender exists, the default is to assume surrender. Target classified as hors de combat pending confirmation.
Intelligence staleness: beyond applicable threshold	Tier L2 and above	Intelligence refresh to within the applicable staleness threshold. Thresholds: HUMINT 4 hours, SIGINT 15 minutes, IMINT 2 hours, GEOINT 24 hours.
CDE confidence interval includes protected persons	Tier L3 and above	Revised strike parameters that reduce the effects radius, or delay until protected persons have moved outside the effects radius at acceptable confidence.
Legal advisor confirmation: not obtained	Tier L3 and above	Legal advisor review and documented compliance determination. Cannot be substituted by any other authoriser regardless of rank.
Active spoofing indicator detected	All lethal tiers	Human intelligence assessment team evaluation of adversarial deception hypothesis. Cannot be cleared by algorithmic counter-spoofing assessment alone.
CDE methodology confidence: below minimum threshold	Tier L3 and above	Additional collateral damage analysis by a qualified human analyst using a validated methodology.

A system that presents a recommendation to an authoriser while one of these Known Unknowns is unresolved is not compliant with EAAP-LGF regardless of what the authoriser then decides. These are architectural requirements, not procedural guidelines.

Appendix B: Illustrative Legal Compliance Assessment Criteria

The following criteria are illustrative of what the Legal Compliance Assessment (Assessment 2 of the Compound Assessment Gate at Layer 0) must evaluate. This list is not exhaustive and does not constitute legal advice. The encodable LCA ruleset must be developed and reviewed by qualified IHL counsel before any operational implementation.

B.1 Military Objective

Criterion	IHL Source	Assessment Question	Governance Consequence
Military objective nature	AP I Art. 52(2)	Does the target by its nature make an effective contribution to military action, and would its neutralisation offer a definite military advantage?	YES: proceed. NO: HARD BLOCK. UNCERTAIN: legal advisor review (LCA Mode B).
Military objective use or purpose	AP I Art. 52(2)	Is the target by its location, purpose, or use making an effective contribution to military action?	Cannot be automated legal advisor review (LCA Mode B mandatory).
Distinction confirmed non-civilian	AP I Art. 48	Is the target clearly and positively distinguished from civilian persons and civilian objects based on confirmed intelligence?	If based solely on pattern-of-life inference: evidence class capped at 25. If unconfirmed: HARD BLOCK.
Presumption of civilian status	AP I Art. 50(1)	Where status is in doubt, is the person being treated as civilian until positive combatant status is confirmed?	Default civilian presumption must be encoded. Uncertainty defaults to civilian at Layer -1.

B.2 Proportionality

Criterion	IHL Source	Assessment Question	Governance Consequence
Proportionality anticipated civilian harm	AP I Art. 51(5)(b)	Is the anticipated incidental civilian harm excessive in relation to the concrete and direct military advantage anticipated?	Automated assessment where CDE is clearly below threshold. Legal advisor review where within defined margin. HARD BLOCK where assessment identifies excessive harm.
Proportionality concrete and direct advantage	AP I Art. 51(5)(b)	Is the anticipated military advantage concrete (specific and identifiable) and direct (close causal relationship to the anticipated attack)?	Speculative or indirect advantage cannot support proportionality. Legal advisor confirmation required where characterisation is contested.
Proportionality assessment timing	AP I Art. 57(2)(a)(iii)	Has the proportionality assessment been conducted before the decision to attack?	Pre-action audit record must exist before Layer 3 authorisation is sought. Assessment written after the decision is architecturally blocked.

B.3 Precautions in Attack

Criterion	IHL Source	Assessment Question	Governance Consequence
Target verification	AP I Art. 57(2)(a)(i)	Has everything feasible been done to verify the target is a military objective and not civilian?	Algorithm-only identification: HARD BLOCK at Layer 1. Single source without cross-cue: HARD BLOCK for Tier L3 at Layer 0.
Choice of means least civilian harm	AP I Art. 57(2)(a)(ii)	Among available means, have those causing least danger to civilian lives been chosen?	Alternative means with materially less civilian harm and feasible:

Criterion	IHL Source	Assessment Question	Governance Consequence
			selection of higher-harm means requires legal advisor justification.
Effective advance warning	AP I Art. 57(2)(c)	Has effective advance warning been given to civilians, unless circumstances do not permit?	If warning is feasible and not given without documented rationale: legal advisor review required.
Attack cancellation changed circumstances	AP I Art. 57(2)(b)	If it becomes apparent that excessive civilian harm will result, will the attack be cancelled or suspended?	Any change in assessment basis after token issuance requires reassessment. Token expiry (15 minutes for L3) is designed to force this check.
Choice of attack objective	AP I Art. 57(3)	When a choice is possible between several objectives achieving similar advantage, has the one expected to cause least civilian danger been selected?	Multi-objective scenarios require documented comparative assessment. Legal advisor review where objectives differ in civilian harm potential.

B.4 Specific Prohibitions

Criterion	IHL Source	Assessment Question	Governance Consequence
Prohibition of perfidy	AP I Art. 37	Does the recommended action involve inviting adversary confidence by feigning protected status, with intent to betray that confidence?	Perfidy is absolutely prohibited. Encoded as a Layer –1 categorical prohibition.
Prohibition of indiscriminate attack area weapon	AP I Art. 51(4)(b)	If an area weapon is involved, can the effects be confined to the specific military objective?	If area effect cannot be confined and civilian presence is confirmed: HARD BLOCK at Layer 0 CDE assessment.
Prohibition of indiscriminate attack method	AP I Art. 51(4)(b)	Is the method of attack one that cannot be directed at a specific military objective?	Indiscriminate methods absolutely prohibited at Layer –1. Layer 0 provides the documented factual basis.
Prohibition of attacks on undefended places	AP I Art. 59, HR Art. 25	If a locality has been declared or clearly appears to be undefended, do the conditions for that declaration remain applicable?	Requires specific legal advisor review. Not amenable to automated determination.

Appendix B provides illustrative criteria only. The criteria reflect well-established IHL principles but are not a substitute for the legal analysis required in specific operational contexts. The LCA ruleset actually implemented must be reviewed and approved by qualified IHL counsel who can assess its application to the specific weapons systems, operational environment, and mandate under which the framework is deployed.